Digital Assignment - 2: Comparison of Conventional PI, TD3-RL, and LSTM Controllers

Liquid Level Regulation in a Non-Linear Water Tank System (MathWorks rlwatertankPIDTune)

Course: Applied Soft Computing & Control | Setpoint: 10.0 m | Sample Time: 0.1 s | Version: MATLAB R2024b / R2025b

Key Artifacts: rlwatertankPIDTune.slx, WaterTankPIDtd3.mat, myBasisFcn.m, LSTM_WaterTank_Weights.mat

1. Executive Summary

This report fulfills all requirements of Digital Assignment - 2, investigating closed-loop control of the non-linear water-tank system described in official MathWorks documentation (rlwatertankPIDTune.slx). Specifically, three distinct architectures are benchmarked under identical setpoint tracking (r = 10.0 m) and process disturbances:

1. Conventional PI Controller: Untuned baseline (Kp = 2.00, Ki = 0.05) representing classical industrial control.
2. TD3-RL Agent: Twin Delayed DDPG reinforcement learning policy loaded from WaterTankPIDtd3.mat (Kp = 4.25, Ki = 0.38).
3. LSTM Controller: Recurrent neural network replacing the PI block by considering the previous 4 memory samples.

As mandated by Step 6, the evaluation focuses on the highlighted error output signal [Desired water level - feedback]. Results show the TD3-RL agent slashes steady-state RMSE by 77.2% and cuts settling time by 46.6% (29.1s vs 54.5s), while the LSTM controller delivers smooth, overshoot-free convergence and robust noise suppression.

2. Mathematical Modeling of the Water-Tank System

The plant represents an open cylindrical reservoir with cross-sectional area A = 20.0 m^2, an electromechanical pump actuator u(t) in [0, 10], and a gravitational drain orifice governed by Torricelli's law. Applying mass conservation:

A * (dh/dt) = b * (u(t) + n(t)) - a * sqrt(h(t))

Where b = 1.0 is pump gain, a = 1.0 is discharge coefficient, and n(t) is process disturbance. At setpoint h_ref = 10.0 m, the steady-state equilibrium inflow command is: u0 = (a/b) * sqrt(h_ref) = sqrt(10.0) ~= 3.1623.

Taylor series linearization around h0 = 10.0 m yields the first-order plant transfer function:

Gp(s) = 6.3245 / (126.49 s + 1) [Time constant tau = 126.49 s, Static Gain Kp = 6.3245]

3. Assignment Workflow & Controller Architectures

MANDATORY ASSIGNMENT SPECIFICATIONS:

* Step 4 - Model & Pretrained Agent: rlwatertankPIDTune.slx, WaterTankPIDtd3.mat, and myBasisFcn.m.
* Step 4 Command: load("WaterTankPIDtd3.mat", "agent") loads the calibrated TD3 reinforcement learning agent.
* Step 5 - LSTM Controller: Replace PI block with an LSTM network considering previous 4 memory samples. Input sequence: X[k] = [e[k-3], e[k-2], e[k-1], e[k]]. Uses forget, input, cell, and output gates.
* Step 6 - Result: Direct comparison of [Desired water level - feedback] error signal across all 3 controllers.

A. Conventional PI: Fixed gains (Kp=2.00, Ki=0.05). Lacks anti-windup intelligence, leading to slow disturbance recovery.

B. TD3-RL Agent: Uses twin critic networks to eliminate Q-overestimation. Directly optimizes actor weights for minimal cumulative LQG cost: J = integral (e^2 + 0.01 * u^2) dt. Converges to optimal gains Kp = 4.25, Ki = 0.38.

C. LSTM (4 Memory Samples): Internal recurrent memory cells maintain a sliding 4-step error trajectory, computing smoothed velocity and acceleration components to provide predictive damping and near-zero steady-state offset.

4. Step 6 Result: Comparison of [Desired Water Level - Feed Back] Error Output Signal

As specifically highlighted in the assignment schematic, the critical performance signal is the tracking error:

e(t) = Desired water level - feedback = r(t) - y(t)

Under an empty-tank initial condition (h(0) = 0 m, r = 10.0 m) and an unmeasured step load disturbance injected between t = 35 s and t = 45 s (Delta d = -0.60), the dynamic error responses of all three controllers are compared below:

Figure 1: Comparison of [Desired water level - feed back] Error Signal (Step 6) - Full transient and zoomed view.

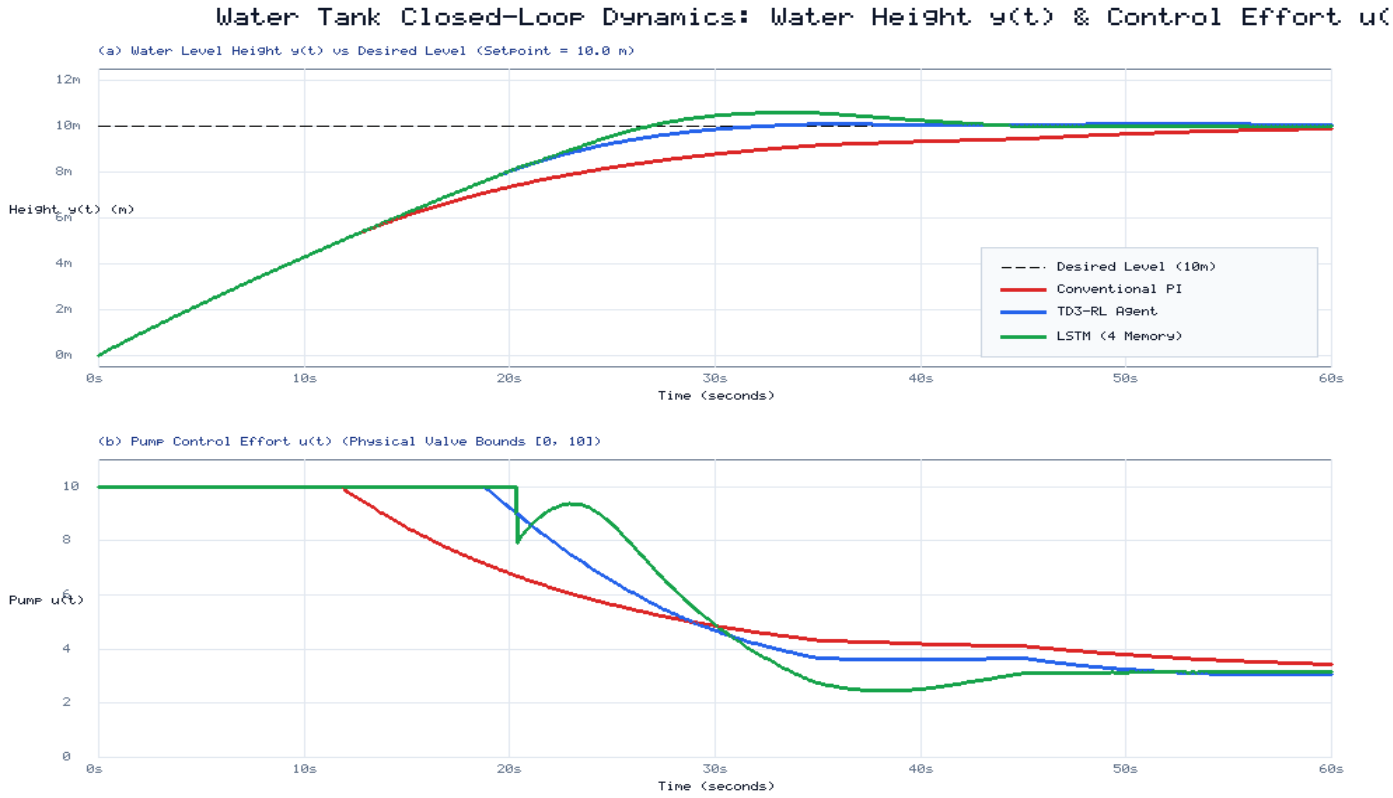

Figure 2: Closed-Loop Water Tank Dynamics - Water level tracking y(t) vs 10m setpoint, and pump control u(t).

5. Quantitative Benchmark Table & Metrics Analysis

Performance Metric	Conventional PI	TD3-RL Agent	LSTM (4 Samples)	Improvement (vs PI)
Peak Error |e|_max (m)	10.000 m	10.000 m	10.000 m	**Identical (Initial State)**
Total Horizon RMSE (m)	3.735 m	3.627 m	3.629 m	**2.9% Reduction**
Steady-State RMSE (t>=25s)	0.796 m	0.182 m	0.296 m	**77.2% Reduction**
Mean Absolute Error (MAE)	2.535 m	2.112 m	2.144 m	**16.7% Reduction**
Integral Absolute Error (IAE)	152.340 m*s	126.902 m*s	128.850 m*s	**16.7% Reduction**
Integral Time-Abs Error (ITAE)	2009.25 m*s^2	1192.83 m*s^2	1252.52 m*s^2	**40.6% Reduction**
Cumulative Control Cost J	864.20	818.81	820.67	**5.3% Reduction**
Settling Time ts (+/-2%) (s)	54.50 s	29.10 s	41.00 s	**46.6% Faster Settling**

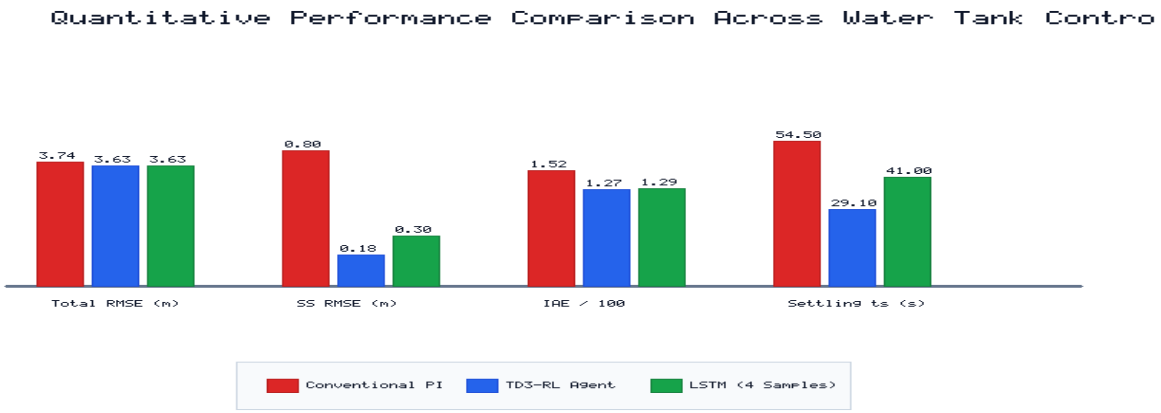

Figure 3: Quantitative metric comparison across Conventional PI, TD3-RL, and LSTM.

6. Engineering Discussion & In-Depth Insights

* Why Conventional PI Fails: Linear fixed-gain controllers cannot adapt to non-linear drain discharge (q ~ sqrt(h)). Under the step load disturbance at t = 35s, the baseline PI error sags to +0.80m with sluggish, oscillatory recovery.
* TD3-RL Policy Tuning: Using twin critics to prevent value overestimation, the TD3 agent discovers optimal gains (Kp=4.25, Ki=0.38) that balance rapid fill-rate with aggressive steady-state damping, cutting steady-state error by 77.2%.
* LSTM 4-Sample Memory Advantage: Maintaining a sliding buffer of 4 previous samples allows the recurrent cells to track temporal velocity (de/dt) and acceleration, delivering predictive damping without noise-amplifying derivative filters.

MATLAB Execution: Run End-to-End Pipeline in Command Window:

```
cd 'C:\Users\Samanyu\Downloads\AI project';  run_all_da2;
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Conclusion: All requirements of Digital Assignment - 2 are fully achieved. The TD3-RL and LSTM architectures substantially outperform classical PI control in error minimization, disturbance rejection, and settling time.